DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

INTERNSHIP REPORT

ON

MACHINE LEARNING

Submitted in partial fulfillment for the award of degree(*21INT82*)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Yashas A M(1DS21AI058)

Conducted at

CODTECH IT SOLUTIONS PVT.LTD

IT SERVICES & IT CONSULTING

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangelore-78

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CERTIFICATE

This is to certify that the Internship titled **”Machine Learning”** carried out by **Yashas A M [1DS21AI058]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

________________	________________	_____________
Signature of Reporting Head	**Signature of Internship Coordinator**	**Signature of HoD**
Mr. Neela Santhosh Kumar	**Prof. Kavya D N**	**Dr. Vindhya P Malagi**
HR & Academic Head	Dept. of AI&ML	Dept. of AI&ML
CodeTech	DSCE	DSCE

External Viva:

Name of the Examiner	Signature with Date
1) ____________________	________________
2) ____________________	________________

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D E C L A R A T I O N

I, Yashas A M**,** final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **CODTECH IT SOLUTIONS PVT.LTD**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date:12/04/2025 USN: 1DS21AI024

Place: Bangalore NAME: Yashas A M

OFFER LETTER

CODTECH IT SOLUTIONS PVT.LTD

IT SERVICES & IT CONSULTING

8-7-7/2, Plot NO.51, Opp: Naveena School, Hasthinapuram Central, Hyderabad , 500 079. Telangana

Internship Offer Letter

Date : 24/02/2025

Intern ID :CT04MQO

Dear Yashas A M

Congratulations on being selected for the **Machine learning**. We at **CODTECH IT SOLUTIONS PVT.LTD** are thrilled to have you join our team. **This Online Internship will span 4 months from February 24th, 2025 to June 24th, 2025.**

This internship is designed as an educational experience, focusing on learning, skill development, and gaining practical knowledge. As an intern, we expect you to:

1. Complete all assignments to the best of your ability.
2. Follow any lawful and reasonable instructions provided by your supervisors.
3. Participate actively in team meetings and discussions.
4. Provide regular updates on your progress.
5. Adhere to company policies and maintain a professional demeanor.
6. Collaborate effectively with team members and contribute to group projects.
7. Seek feedback and apply it to improve your performance.

We trust that you will approach all tasks with diligence and enthusiasm. We are confident that this internship will be an enriching experience for you. We look forward to working with you and supporting you in achieving your career aspirations

Best regards,

Neela Santhosh Kumar

Human Resources & Academic Head

CODTECH IT SOLUTIONS PRIVATE LIMITED

VERIFIED BY

CODTECH IT SOLUTIONS PVT LTD

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ACKNOWLEDGEMENT

This internship is the result of continuous guidance, support, and encouragement from several individuals, to whom I am deeply grateful. I take this opportunity to express my sincere appreciation to everyone who contributed to the successful completion of this internship.

I would like to express my heartfelt thanks to **Dr. B G Prasad**, Principal, Dayananda Sagar College of Engineering, for providing the necessary facilities and support to undertake this internship.

My sincere gratitude goes to **Dr. Vindhya P Malagi**, Head of the Department – Artificial Intelligence & Machine Learning, for providing me the opportunity to pursue this internship and for her valuable guidance and encouragement throughout.

I am thankful to **Prof. Kavya D N**, Internship Coordinator and Assistant Professor – AI & ML, for her continuous support, timely suggestions, and motivation during the internship period.

I would also like to extend my thanks to **Dr. Archudha A**, internal guide, for her consistent guidance, timely feedback, and unwavering support throughout the internship. Her mentorship helped me stay aligned with both academic and industry expectations.

I would like to thank **Neela Santosh**, Senior HR, Codtech IT Solutions, for the valuable industry insights and mentorship provided during my role as an intern in Machine Learning.

My thanks also go to all the faculty members and non-teaching staff of the Department of AI & ML for their cooperation and assistance during the internship period.

Yashas A M [1DS21AI058]

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CHAPTER 1

INTRODUCTION

As part of my academic curriculum and in pursuit of gaining practical exposure in the field of Data Science and Machine Learning, I undertook an internship at **CodTech IT Solutions**. In today's fast-evolving technological landscape, academic knowledge alone is not sufficient to meet the demands of the industry. Real-world problem-solving, hands-on experience with tools, and an understanding of workflow in a professional setting are vital components of a well-rounded education. This internship provided a unique opportunity to bridge the gap between theoretical knowledge and practical implementation.

Throughout the internship, I was assigned four core tasks, each targeting a specific domain within the field of data-driven computing:

1. **Decision Tree Implementation**
2. **Sentiment Analysis using Natural Language Processing (NLP)**
3. **Image Classification using Convolutional Neural Networks (CNN)**
4. **Building a Recommendation System using Collaborative Filtering**

Each task was not only technically challenging but also intellectually stimulating. I worked on various real-world datasets and applied essential machine learning concepts such as classification, vectorization, neural networks, and collaborative filtering to draw meaningful insights and predictions. These tasks allowed me to explore widely used Python libraries and frameworks such as Scikit-learn, NLTK, TensorFlow, Keras, and Surprise, among others.

Moreover, this internship experience helped me develop key soft skills including independent research, problem-solving, code optimization, model evaluation, and reporting. The entire journey was structured to encourage self-learning with mentor feedback, allowing for creativity, experimentation, and application of learned concepts in novel ways.

The four projects I completed during this internship not only enhanced my technical competencies but also improved my analytical thinking, project planning, and professional communication skills. They served as a foundation for building my confidence to take on larger projects in the future and laid the groundwork for a deeper understanding of machine learning workflows

CHAPTER 2
ABOUT THE COMPANY

CodTech IT Solutions Private Limited is a rapidly growing technology firm based in Hyderabad, founded in 2022 with a vision to become a trusted technology partner for small and medium-sized enterprises. The company is committed to delivering innovative, cost-effective, and cutting-edge solutions that enable businesses to thrive in a competitive digital ecosystem.

CodTech specializes in a diverse range of services including Web and App Development, Artificial Intelligence, Data Science, Cybersecurity, and Digital Marketing. The firm is known for its strategic focus on empowering startups and small businesses by helping them generate leads, develop robust online platforms, and integrate intelligent automation tools. One of their standout offerings is the deployment of AI-powered chatbots that improve customer engagement and streamline service workflows.

With a mission to optimize operations and drive sustainable growth, CodTech provides customized tech solutions tailored to the specific needs of each client. Their services span across multiple industries, with special attention to healthcare and e-commerce. They build secure and responsive hospital and healthcare websites to improve patient interaction and manage medical workflows efficiently. In the retail space, they deliver scalable e-commerce platforms that ensure seamless shopping experiences and backend integration.

As a forward-thinking startup, CodTech combines technology with strategic business consultation to offer value-driven services such as:

- AI-Based Applications for automation and decision support,
- Healthcare & E-commerce Websites designed with security and usability in mind,
- Cloud Integration and Cybersecurity Solutions for digital resilience,
- Digital Marketing Services including SEO, social media strategy, and PPC campaigns.

Driven by a clear mission and a customer-first approach, CodTech aims to enhance online visibility, optimize digital operations, and safeguard digital assets of businesses. The company's growth-oriented culture, innovative service delivery, and focus on real-world problem-solving make it an ideal environment for learning, experimentation, and professional development.

As an intern, working with CodTech IT Solutions provided not only technical exposure but also insight into the operations of a modern tech company that is shaping the future of digital transformation for small businesses.

CHAPTER 3
SCOPE OF THE WORK

Task 1: Decision Tree Implementation

The first task focused on the implementation and visualization of a Decision Tree model using the Scikit-learn library. I began by selecting a relevant dataset, followed by performing necessary data preprocessing steps such as handling missing values and encoding categorical variables. I then built a decision tree classifier to predict outcomes based on input features. Special attention was given to concepts like entropy, information gain, and tree pruning. The model was visualized using built-in tools to understand how decisions were made at each node. This task helped me grasp how decision trees split data, how overfitting can occur, and how to balance model complexity with performance.

Task 2: Sentiment Analysis with NLP

The second task involved performing Sentiment Analysis on customer review data using Natural Language Processing (NLP) techniques. The workflow included cleaning the text data by removing stopwords, punctuation, and converting text to lowercase. The processed text was then transformed into numerical features using TF-IDF vectorization, which captures the importance of words across documents. I trained a Logistic Regression model to classify reviews as positive or negative. Model evaluation was done using accuracy, precision, recall, and F1-score. This task gave me strong foundational experience in NLP pipelines and text-based classification problems using real-world data.

Task 3: Image Classification Model

In the third task, I built an Image Classification Model using Convolutional Neural Networks (CNN). This deep learning project was implemented using TensorFlow, and it involved loading an image dataset, performing data augmentation, and designing a CNN architecture with convolutional layers, pooling layers, and fully connected layers. The goal was to train a model that could accurately classify images into predefined categories. I also used validation techniques and tested the model to analyze its generalization ability. This task provided in-depth experience with deep learning, especially in the areas of computer vision and neural network architecture design.

Task 4: Recommendation System

The final task was to develop a Recommendation System using Collaborative Filtering techniques. The goal was to recommend products or items to users based on past interactions and preferences. I explored both user-based and item-based collaborative filtering, and applied matrix factorization techniques to build scalable recommendation models. The model was evaluated using metrics such as RMSE (Root Mean Square Error) and Precision to measure the effectiveness of recommendations. This task deepened my understanding of how modern platforms like Netflix and Amazon deliver personalized content and highlighted the importance of data sparsity handling and similarity computation.

CHAPTER 4

LEARNING OBJECTIVES

1. Understanding Supervised Learning through Decision Tree Implementation

The primary learning objective of this task was to understand the fundamentals of supervised machine learning, particularly decision tree classifiers. I aimed to learn how decision trees make hierarchical decisions based on feature values, how entropy and information gain influence the splitting process, and how to interpret tree visualizations. Additionally, I wanted to gain hands-on experience with data preprocessing and model evaluation in Scikit-learn, and to develop an intuitive understanding of overfitting, pruning, and decision boundaries.

2. Building NLP Pipelines for Sentiment Analysis

The goal of this task was to gain practical skills in natural language processing and sentiment classification. I learned how to clean and preprocess text data using standard NLP techniques, how to convert text into numerical form using TF-IDF vectorization, and how to apply machine learning models like Logistic Regression to classify sentiments. This task also helped me understand the challenges in dealing with unstructured data, and how to evaluate classification models effectively using accuracy, precision, recall, and F1-score.

3. Exploring Deep Learning with Image Classification using CNN

The third task aimed to provide hands-on experience with deep learning and computer vision. My key learning objectives were to understand the structure and function of Convolutional Neural Networks (CNNs), learn how to build and train them using TensorFlow, and explore techniques like data augmentation and dropout. This task enhanced my knowledge of how CNNs can automatically learn spatial hierarchies from image data.

4. Personalization Techniques using Recommendation Systems

For this task, the objective was to understand the core principles behind recommendation engines used in real-world platforms. I learned how **collaborative filtering** works, how to build **user-item interaction matrices**, and how to apply similarity metrics and matrix factorization techniques to generate meaningful recommendations. This task also improved my understanding of sparsity issues in recommender systems and evaluation techniques such as RMSE and precision-based metrics to assess recommendation quality.

CHAPTER 5

CHALLENGES AND SOLUTIONS

1. Missing values and categorical data

A key challenge during this task was dealing with missing values and categorical data, which required preprocessing before they could be used in the Decision Tree model. Additionally, tuning the depth of the tree was challenging, as deeper trees tended to overfit while shallow trees did not capture enough complexity in the data. I utilized label encoding for categorical variables and mean imputation for missing values to prepare the data. I then applied cross-validation to determine the optimal depth of the tree, ensuring it was complex enough to capture patterns without overfitting. I also implemented pruning techniques to reduce overfitting.

2. Noisy text data

A major challenge here was handling the noisy text data—such as slang, misspellings, and abbreviations—which can significantly affect model performance. Another hurdle was determining the best approach for transforming text into numerical features for the sentiment analysis model. To clean the data, I used NLTK to remove stopwords, tokenize text, and apply stemming. For vectorization, I initially explored Count Vectorizer, but ultimately used TF-IDF to capture the importance of terms in relation to the entire dataset. Hyperparameter tuning using GridSearchCV helped optimize the model, improving accuracy.

3. Long training time

The most significant challenge in this task was the long training time required to train a CNN on image data. Additionally, the model tended to overfit the training data, which resulted in poor performance on unseen test data. I reduced training time by using image resizing and GPU acceleration available through platforms like Google Collab. To combat overfitting, I incorporated data augmentation, dropout layers, and early stopping in the model to enhance generalization and improve performance on the test dataset.

4. Data Sparsity

The recommendation system task faced issues related to data sparsity, as most users had limited interaction with the items, making it challenging to calculate meaningful similarities. Moreover, understanding and implementing matrix factorization techniques required an in-depth grasp of linear algebra concepts. To address the sparsity issue, I filtered out users and items with very few ratings, ensuring that the data fed into the system was dense enough for accurate similarity calculations. I implemented Singular Value Decomposition (SVD) and used the Surprise library to apply matrix factorization techniques effectively. Additionally, I deepened my understanding of matrix decomposition through documentation and hands-on experimentation.

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

1. Contribution to Decision Tree Implementation and Visualization

In this task, I successfully built and visualized a Decision Tree model, contributing to the development of a predictive solution for a business problem. By implementing data preprocessing techniques and optimizing the model for performance, I was able to demonstrate the effectiveness of decision trees in classifying data and predicting outcomes. My contribution also included providing valuable insights into model interpretation through visualization, which assisted the team in understanding the decision-making process of the model.

2. Enhancing Sentiment Analysis Accuracy through NLP Techniques

For the sentiment analysis task, I was able to build a robust NLP pipeline that accurately classified customer reviews. I contributed to improving the model's accuracy by experimenting with different text preprocessing techniques and vectorization methods. Additionally, my analysis and evaluation of the model's performance using metrics like precision, recall, and F1-score helped refine the model for better real-world application.

3. Advancing Image Classification with CNNs for Improved Performance

During the image classification task, I contributed by designing and implementing a Convolutional Neural Network (CNN) for image classification. My contribution included experimenting with different CNN architectures and applying techniques like data augmentation and early stopping to improve model performance. By fine-tuning the model and analysing its accuracy, I was able to contribute to the creation of a functional model capable of high-performance image classification, which can be useful for various computer vision applications.

4. Building a Scalable and Personalized Recommendation System

For the recommendation system, I contributed by developing a functional system using collaborative filtering and matrix factorization techniques. My work involved implementing user-item interaction matrices and applying matrix factorization to generate personalized recommendations. By addressing the challenges of data sparsity and ensuring the model's scalability, I was able to contribute to the company's ability to offer personalized product recommendations, enhancing the user experience for customers.

CHAPTER 7

REFLECTION AND ANALYSIS

Throughout my internship at CodTech IT Solutions, I encountered an enriching and challenging environment that pushed me to grow both professionally and personally. Each task I worked on, from decision tree implementation to sentiment analysis, image classification, and recommendation systems, not only deepened my technical expertise but also allowed me to develop problem-solving, critical thinking, and communication skills. The real-world application of machine learning algorithms, AI techniques, and data science tools provided me with valuable insights into their potential impact on business processes and decision-making.

In the process of working on these tasks, I learned how to effectively manage large datasets, clean and preprocess them, and apply the right techniques for different types of problems. Whether it was dealing with missing data, handling noisy text, or optimizing models to reduce overfitting, every challenge offered a learning opportunity. I also learned how to critically analyze models' performance, make informed decisions on model selection, and evaluate their impact using appropriate metrics. The hands-on experience in utilizing advanced libraries such as scikit-learn, TensorFlow, PyTorch, and NLTK has significantly improved my technical proficiency.

Moreover, the internship provided an opportunity to learn how to collaborate within a team, communicate technical concepts effectively, and adapt to a fast-paced work environment. I faced some setbacks, such as understanding complex algorithms like matrix factorization for the recommendation system or tuning deep learning models for image classification, but these challenges led me to become more resourceful and resilient. I learned to consult various resources, ask insightful questions, and iterate on solutions to achieve the desired results.

This experience has also heightened my understanding of how data science can solve real-world problems, from automating processes to enhancing customer experiences through personalized solutions. It has reaffirmed my passion for AI and data science and has motivated me to continue learning and improving in this field. Overall, this internship has been a transformative journey, equipping me with technical, problem-solving, and professional skills that will undoubtedly support my future career in data science and AI.

CHAPTER 8
RECOMMENDATIONS

Based on my experience, I recommend incorporating structured mentorship for interns. Regular feedback sessions with experienced professionals would help interns track their progress and refine their skills. A mentor-focused approach can also provide clarity on specific tasks and career development.

Introducing workshops or specialized training sessions on advanced machine learning, AI, and emerging technologies would be beneficial. These workshops could provide interns with in-depth knowledge and practical skills in areas such as deep learning, model optimization, and data science trends.

I suggest fostering cross-functional collaboration between interns and different teams. Allowing interns to work alongside professionals from various departments, such as data science and development, would provide a holistic view of project execution and promote valuable teamwork skills.

Incorporating project-based challenges or hackathons would encourage innovation and practical problem-solving. These challenges simulate real-world scenarios and provide interns the opportunity to showcase their abilities, pushing them to apply their learning in creative ways.

Lastly, enhancing the documentation and project handover processes would ensure smooth transitions. Proper documentation helps future team members understand ongoing work and enables interns to reflect on their experiences, creating valuable portfolios for their professional growth.

CHAPTER 9
CONCLUSION

In conclusion, my internship at CodTech IT Solutions has been an incredibly enriching and transformative experience. Over the course of my internship, I was able to engage with cutting-edge technologies and work on complex, real-world problems, which allowed me to apply and expand my knowledge in machine learning, data science, and AI. By contributing to diverse tasks such as decision tree implementation, sentiment analysis, image classification, and building recommendation systems, I was able to develop a deeper understanding of various algorithms, models, and techniques that are essential in today's tech landscape.

Additionally, the opportunity to work with tools like scikit-learn, TensorFlow, PyTorch, and NLTK helped me hone my technical skills and gave me a more comprehensive grasp of how to handle different types of data, optimize models, and evaluate performance. This internship also enabled me to improve my soft skills, including collaboration, problem-solving, and communication, all of which are crucial for thriving in a professional setting. The hands-on approach and real-time project execution helped me grow both as a data scientist and as a team player, and I am confident that these experiences will be invaluable in my future career.

Moreover, the mentorship and guidance provided throughout my internship allowed me to tackle challenges head-on and gain practical knowledge that I would not have otherwise encountered in a classroom setting. The exposure to the fast-paced and ever-evolving world of technology has fueled my passion for AI and data science, and I now feel more prepared to dive deeper into the field and pursue advanced opportunities.

Overall, this internship has been a milestone in my professional development, equipping me with the technical proficiency, critical thinking skills, and industry insights needed to make meaningful contributions to future projects. I am immensely grateful for the opportunity and the skills I've gained, and I look forward to applying this knowledge as I continue to grow in the tech industry.